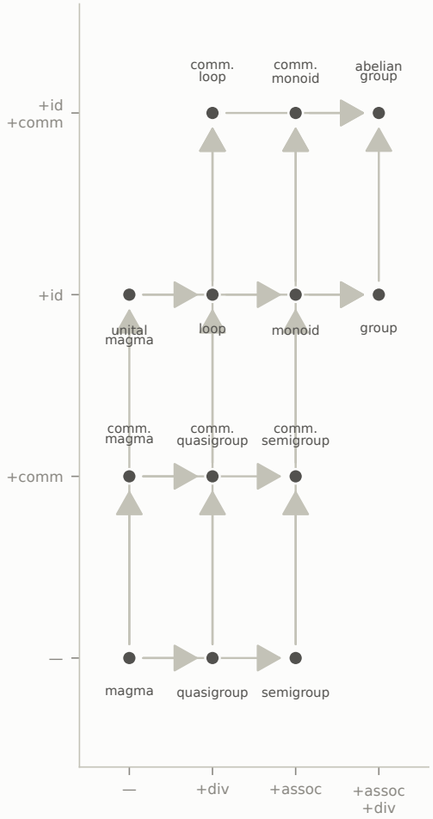
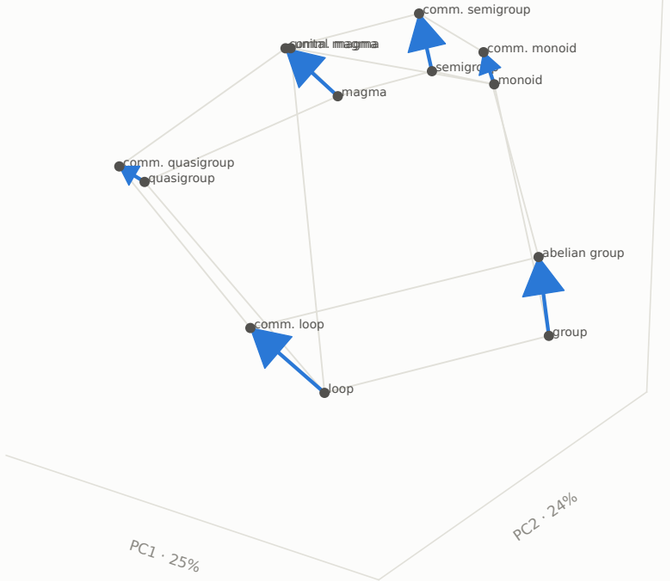
1 · Structures are sets of axioms



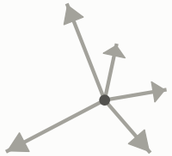
2 · Adding an axiom is an arrow + commutativity in Qwen3.5-2B (top-3 PCs)



3 · Line the arrows up

consistency 0.53

random-init twin

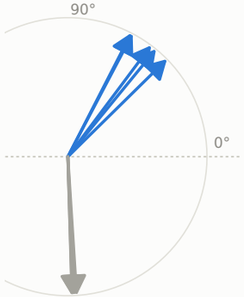


consistency 0.01

4 · Measure in the full space: angle of each arrow to the others' mean

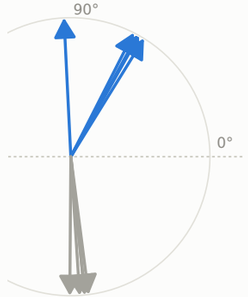
model above the dashed line (blue), twin below (grey); 0° = same direction, 90° = unrelated

+ associativity



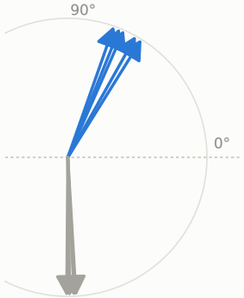
consistency 0.43 (p = 0.003)
twin 0.02

+ identity



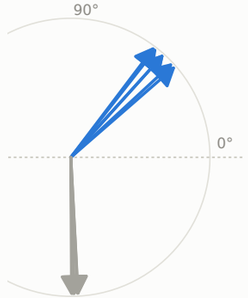
consistency 0.24 (p = 0.042)
twin 0.04

+ divisibility (inverses)



consistency 0.29 (p = 0.027)
twin 0.01

+ commutativity



consistency 0.53 (p < 0.001)
twin 0.01